DECODELABS | DATA ANALYTICS

PROJECT 1
CHANGE LOG

Data Cleaning & Preparation

Organization	DecodeLabs
Dataset	Dataset for Data Analytics.xlsx
Dataset size	1,200 rows × 14 columns
Tools	Python, Pandas, VS Code, Excel / openpyxl
Prepared by	[Enter your full name]
Date	[Enter submission date]

Purpose: This change log records the data-quality checks and cleaning actions performed on the supplied dataset, along with the reported before-and-after results.

1. Summary of Changes

Area	Initial finding	Action taken / decision	Final result
Missing values – CouponCode	309 missing values	Filled missing entries with “No Coupon”; records were retained.	0 missing values
Missing values – other columns	0 missing values	No action required.	0 missing values
Completely duplicated rows	0 duplicates	No rows removed because no complete duplicate rows were found.	0 duplicates
Duplicate OrderID values	0 duplicate IDs	No ID records removed; uniqueness check retained for validation.	0 duplicate IDs
Date values	0 invalid dates reported	Date field converted/validated using Pandas datetime handling.	0 invalid dates; dtype datetime64[us]
Numeric values	No negative values reported	Quantity, UnitPrice, ItemsInCart and TotalPrice checked.	No negative values reported
TotalPrice calculation	Checked against Quantity × UnitPrice	Compared after rounding both calculated and actual totals to 2 decimal places.	1,200 matched; 0 mismatched

2. Detailed Cleaning Notes

• **CouponCode:** The 309 blank coupon entries were replaced with the text “No Coupon”. This preserved the order records while making the absence of a coupon explicit.

• **Duplicate checks:** The dataset contained zero fully duplicated rows and zero duplicate OrderID values. No rows were deleted for duplication. Repeated CustomerID values were not treated as duplicates because a customer can place more than one order.

• **Date validation:** The Date field was processed with Pandas datetime conversion/validation. The reported final check found no invalid dates.

• **Numeric validation:** Quantity, UnitPrice, ItemsInCart and TotalPrice were checked. TotalPrice was validated against Quantity multiplied by UnitPrice, with both values rounded to two decimal places before comparison. All 1,200 records matched.

3. Overall Before-and-After Result

Metric	Before	After
Rows	1,200	1,200
Columns	14	14
Missing values	309	0
Duplicate OrderIDs	0	0
Invalid dates	0	0
TotalPrice mismatches	0 reported	0

4. Export and Evidence Status

The cleaned workbook is expected to be saved as **cleaned_dataset.xlsx** in the **Cleaned_Data** folder. The export success message and the existence of the saved workbook must be confirmed before final submission. Add the three execution screenshots to the Project Report: (1) missing values before and after, (2) duplicate ID/date/TotalPrice validation, and (3) successful export and saved file path.

5. Final Review Checklist

■ Replace [Enter your full name] and [Enter submission date].

■ Confirm cleaned_dataset.xlsx exists in the Cleaned_Data folder.

■ Confirm the Python script is saved in the Python_code folder.

■ Insert the required screenshots into Project_Report.docx.

■ Save this PDF as Change_Log.pdf in the Documentation folder.

Note: This change log reflects the cleaning results and program outputs reported during the project work. It does not claim that the Excel export has been completed until the saved file is checked.